LING285-L2

Course: LING285: Human Language and Technology
Title of Lecture: Lecture 2 - Phonetics and AI
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Section 1: Watching Speech Inside the Vocal Tract

- **Real-time MRI** (magnetic resonance imaging) lets us film the inside of the head while a person is talking. The images are **midsagittal**, which means they are a side-view slice taken straight down the middle of the head, from the nose to the back of the neck. The face points to the left, so the lips are on the left edge and the spinal column is the bright vertical band on the right. The recordings run for about half a minute each and include the sound of the speaker, so every movement can be matched to the speech it produces.
    - In MRI, **air shows up black** because it gives off almost no signal, and soft tissue like the tongue shows up gray or white. The dark spaces in the image are therefore the open airways: the mouth cavity, the nasal passages above the roof of the mouth, and the **pharynx**, which is the air tube at the back of the throat behind the tongue.
    - The large rounded gray mass filling most of the mouth is the **tongue**. Above it is the **hard palate**, the bony roof of the mouth, and hanging down from the back of the hard palate is the **velum**, or soft palate, which can be raised or lowered to close or open the path from the throat into the nose.
- When these recordings play, the tongue, lips, jaw, and velum are all moving at once and very quickly. Every consonant and vowel we hear is the result of the speaker shaping this tube of air in a particular way. **Phonetics** is the study of these speech sounds, and the images give us a direct way to see the shapes behind them.

Section 2: Fricatives

- A **fricative** is a consonant made by bringing two parts of the mouth very close together without closing them completely. Air is pushed through that narrow gap, and the squeezed air becomes turbulent, which produces the hissing or buzzing noise we hear in sounds like f and s. A fricative can be held for as long as we have breath, which is different from sounds that require a full closure.
- English fricatives are organized along two dimensions: **place of articulation** and **voicing**. Place of articulation is where in the mouth the narrow gap is formed. Voicing is whether the **vocal folds** in the larynx are vibrating while the sound is made.
    - A **voiceless** sound is made with the vocal folds apart and not vibrating, so the only noise is the hiss of air through the gap.
    - A **voiced** sound is made with the vocal folds vibrating, so the hiss is combined with a buzz. Placing a hand on the throat while saying "ssss" and then "zzzz" makes the difference easy to feel, since only the second produces vibration.
- The English fricatives fall into **four places of articulation**, and each place has a voiceless and a voiced member.
    - **Labial** fricatives are f (voiceless) and v (voiced). The word labial refers to the lips, and for f and v the lower lip is raised up against the upper front teeth.
    - **Dental** fricatives are both spelled th. The voiceless one is the sound at the start of "think," and the voiced one is the sound at the start of "this." The tip of the tongue is placed against or just between the upper teeth.
    - **Alveolar** fricatives are s (voiceless) and z (voiced). The **alveolar ridge** is the bony ridge just behind the upper front teeth, and the tongue tip is raised close to it.
    - **Postalveolar** fricatives are sh (voiceless) and the voiced partner symbolized as j, which is the sound in the middle of "measure" and "vision." The narrow gap is formed just behind the alveolar ridge, a little farther back in the mouth than for s and z.

- The four MRI images line up with these four places, and they show how the constriction moves backward from one place to the next.
    - In the labial image, the lower lip is pulled up toward the upper teeth while the tongue stays low and relaxed in the mouth.
    - In the dental image, the front of the tongue is pushed forward toward the teeth.
    - In the alveolar image, the tongue tip is raised up toward the ridge behind the upper teeth.
    - In the postalveolar image, the narrowing sits slightly farther back, with more of the front part of the tongue raised toward the roof of the mouth.
- The pairing matters: f and v, the two th sounds, s and z, and sh and its voiced partner are each made with the **same place and the same manner**. The only thing that separates the members of each pair is whether the vocal folds are vibrating.

Section 3: Stops

- A **stop** is a consonant made by closing the vocal tract completely at some point so that air cannot get out through the mouth. Air pressure builds up behind the closure, and when the closure is released, the trapped air bursts out. English has two kinds of stops, **oral stops** and **nasal stops**.
    - In an **oral stop**, also called a **plosive**, the velum is raised and presses against the back wall of the throat, which seals off the nose. With both the mouth and the nose closed, pressure builds and is released as a small burst of air.
    - In a **nasal stop**, the mouth is closed in exactly the same way, but the velum is lowered. Air flows up and out through the nose the whole time, so there is no pressure build-up, and the sound has a humming quality.
- The stops are grouped into **three places of articulation**, and each place has a voiceless oral stop, a voiced oral stop, and a nasal stop.
    - At the **lips**, p is voiceless, b is voiced, and m is the nasal. The MRI images for p, b, and m all show the two lips pressed together, closing off the front of the mouth.
    - At the **alveolar ridge**, t is voiceless, d is voiced, and n is the nasal. The images for t, d, and n show the tip of the tongue raised and pressed against the ridge behind the upper teeth.
    - At the **velum**, k is voiceless, g is voiced, and ng is the nasal, which is the sound at the end of "sing." A sound made at the velum is called **velar**. The images for k, g, and ng show the back of the tongue humped up and pressed against the soft palate.
- The nasal images differ from the oral images in the position of the velum. In m, n, and ng, the velum hangs down away from the back of the throat, leaving an open passage into the nose, while in p, t, and k the velum is raised to seal that passage. The short video clips of each stop show the closure forming and releasing in real time, which makes it clear that a stop is an event in time: first closure, then pressure or nasal airflow, then release.

Section 4: The Structure of English Consonants

- The fricatives and stops fit together into one table built on three properties: **manner of articulation**, **place of articulation**, and **voice**. Manner of articulation is how the air is shaped, whether through a narrow gap as in a fricative, a full closure with the nose sealed as in an oral stop, or a full closure with the nose open as in a nasal stop.
    - The places run from the front of the mouth to the back: **labial**, **dental**, **alveolar**, **postalveolar**, and **velar**.
    - The fricatives fill the labial, dental, alveolar, and postalveolar columns, each with a voiceless and voiced member: f and v, th and th, s and z, and sh and j.
    - The oral stops fill the labial, alveolar, and velar columns: voiceless p, t, and k, and voiced b, d, and g.
    - The nasal stops fill the same three columns as the oral stops, and all of them are voiced: m, n, and ng.
- Every one of these consonants can be described by a combination of three values. For example, t is a **voiceless alveolar oral stop**, and z is a **voiced alveolar fricative**. Changing just one value produces a different sound: changing t from voiceless to voiced gives d, and changing d from an oral stop to a nasal stop gives n.

- The empty cells are also informative. None of these English consonants is a dental or postalveolar stop, and none is a velar fricative. The sounds of English are therefore not a random collection of noises but a **highly structured system** of shared properties.
- Every English speaker already uses this system every time they talk, without ever having been taught the table. This means a speaker's knowledge of English contains a great deal of hidden structure, organized along place, manner, and voice.

Section 5: What AI Is

- **Artificial intelligence (AI)** is a model of several **psychological skills that animals have**. The five skills are **knowing**, **recognizing**, **attending**, **acting**, and **learning**.
    - Knowing is having information stored in a usable form.
    - Recognizing is identifying something from the input that arrives.
    - Attending is selecting which parts of the input to focus on.
    - Acting is producing a behavior or output.
    - Learning is changing over time based on experience.
- The way AI models each of these skills is **at least slightly based on how humans perform them**. This goes against a common view that AI is purely a technical innovation that appeared out of nowhere. AI grew out of ideas about human and animal psychology, and each part of an AI system can be traced back to a human skill it was built to imitate.

Section 6: Knowing and Taylor's Cloze Procedure

- The question of how AI **knows** something connects to a 1953 paper by **Wilson L. Taylor** titled **"Cloze Procedure": A New Tool for Measuring Readability**. It was published in the Fall 1953 issue of **Journalism Quarterly**, a journal devoted to research studies in the field of mass communications.
    - The paper was the first comprehensive statement of a research method and its theory, which had been introduced briefly during a workshop at the **1953 AEJ convention**.
    - It reported findings from **three pilot studies and two experiments** in which cloze procedure results were compared with the results of two existing **readability formulas**, which are formulas that estimate how easy a piece of writing is to read.
- Taylor described **cloze procedure** as a new **psychological tool for measuring the effectiveness of communication**. The method is straightforward, the data are easily turned into numbers, and the findings held up in testing.
    - At first the tool was treated as a new approach to **readability**, and it was used that way in the three pilot studies and two experiments.
    - Its results were repeatedly shown to agree with two standard devices for estimating readability, the **Flesch** and **Dale-Chall** formulas. After that, cloze procedure was tested directly against those standard formulas.
    - Taylor expected that if future research supported these results, the tool would have many theoretical and practical uses in other fields that involve communication.
- Taylor credited **Prof. Charles E. Osgood** of the University of Illinois and **Melvin R. Marks** of the Personnel Research Section, A.G.O., Department of the Army, for starting and helping with the work that produced the idea of cloze procedure. Both were **experimental psychologists**.
    - Others who advised or helped included **Prof. Lee J. Cronbach**, an educational psychologist and statistician; **Dean Wilbur Schramm** of the Division of Communications; **Prof. Charles E. Swanson** of the Institute of Communications Research; and **George R. Klare**, a psychologist. Both Swanson and Klare had written articles on readability.
    - **Kalmer E. Stordahl** and **Clifford M. Christensen**, research associates of the Institute, also contributed, along with several journalism teachers who let their classes take part.
- The origin of cloze procedure in experimental psychology is important for AI. A tool built to measure human understanding of language ended up describing what it means for a system to know something.

Section 7: Closure and the Cloze Unit

- At the center of Taylor's method is a unit of measurement called a **cloze**. It is pronounced like the verb "close," and the name comes from the word **closure**.
    - **Closure** is a term from **Gestalt psychology**, the branch of psychology that studies how we perceive whole patterns. It refers to the human tendency to **complete a familiar pattern that is not quite finished**.
    - The standard example is a broken circle. We tend to see a circle with a gap as a whole circle, filling in the missing piece in our minds.
- The black shapes next to Taylor's text show closure in action. There are three black circles, each with a wedge cut out of it, and the wedges all point inward toward a common center. Nothing in the picture is actually a triangle, yet we see a bright white triangle sitting on top of the three circles, with edges that seem to run between the cut-out wedges. The visual system completes the triangle from the partial information the wedges provide.
- Taylor explained that a person can complete a broken circle because its shape is **so familiar** that it can be recognized even when much of it is missing. The same principle applies to language.
    - Given the sentence "Chickens cackle and ___ quack," almost anyone can instantly supply the word "ducks."
    - If the word a person supplies is the same as the word that was removed, that person scores **one cloze unit** for correctly closing the gap in the language pattern.
- Filling that single blank requires a surprising amount of knowledge. The sentence is a complex pattern built out of many smaller patterns.
    - A reader must know the **meanings** of all five words, which Taylor describes as patterns of relationships between symbols and meanings.
    - A reader must also know their **forms**, meaning the patterns of letters that make up each word.
    - Beyond single words, a reader must know the meanings of **combinations** of words.
    - The reader must also notice that the sentence is built in two parallel halves. The first half pairs an animal with its sound, "chickens cackle," so the second half calls for the animal that goes with the sound "quack."
    - In other words, a reader has to guess what the incomplete sentence means as a whole and then complete its pattern to fit that whole meaning.
- Taylor gave two formal definitions.
    - A **cloze unit** is any single occurrence of a **successful attempt to reproduce accurately a part deleted from a "message"**, where a message is any language product, by deciding from the context that remains what the missing part should be.
    - **Cloze procedure** is a method of **intercepting a message from a "transmitter"**, which is a writer or speaker, **mutilating its language patterns by deleting parts**, and giving the damaged message to **"receivers"**, who are readers or listeners. The receivers' attempts to make the patterns whole again produce a large number of cloze units, which can be counted.

Section 8: How AI Knows

- AI shows that it knows something by **demonstrating the ability to complete, or cloze, sequences**. A system that can reliably fill in the missing parts of language has captured the patterns of that language.
- The same kind of test works for people. A teacher can test a student's knowledge of a skill by asking the student to complete some sequence, such as finishing a sentence, supplying the next step in a procedure, or filling in a missing word. Success at completion is taken as evidence of knowledge.
- AI takes this idea to its limit. From the AI point of view, **knowledge is nothing more than completion**. A system counts as knowing English if, given part of an English sequence, it can correctly produce what comes next or what is missing, and no further ingredient beyond that ability is required.

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Section 9: How GPT Represents Words

- **GPT** is a system that takes in a sequence of words and predicts what word should come next. The words themselves are the input, so the first question is what a word looks like to a computer.
- To a computer, a word is simply **a bunch of random numbers**. Each word in the vocabulary is assigned its own set of numbers, and the only strict requirement at the start is that **no two words share the same set of numbers**. The numbers do not need to mean anything at first, as long as each word can be told apart from every other word.
- A list of numbers like this is called a **vector**. The vector for a single word could contain **1024 numbers**.
    - In the diagram, the words "chickens," "cackle," "and," and "ducks" are each drawn as a tall vertical bar. Each bar stands for one vector, a column of 1024 numbers that represents that word.
    - The input to GPT is therefore not letters or sounds but a row of vectors, one for each word in the sequence.
- Given the sequence "chickens cackle and ducks ___," a system that **knows English** will produce an output vector of **10,000 numbers**. Each of those numbers is the **probability that a particular word of English could be the next word**.
    - In the diagram, this output is drawn as one long horizontal bar, with labels for a few of the words it covers: "a," "cat," "quack," and "Zulu."
    - For this sequence, a system that knows English assigns a high probability to "quack," since ducks are the animal that quacks. Words like "cat" and "Zulu" receive very low probabilities, since they do not make sense in that position.
    - The output does not choose a single word directly. It gives a probability for every word in the vocabulary at once, and the most likely words are the ones the system treats as the best completions.
- This connects directly to cloze procedure. The system is being asked to fill in a blank at the end of a sequence, and its knowledge of English shows up entirely in how well it completes that blank.

Section 10: How GPT Transforms Inputs into Predictions

- The list of probabilities that GPT produces is called a **pdf**, short for **probability distribution function**. It describes how likely each possible next word is. The input vectors are labeled the **x's**, so the central question is how the system turns the x's into the pdf.
- GPT transforms the inputs **through recognizing by computing**. It recognizes the pattern in the input words, and that recognition is carried out entirely through computation.
- The computing involved is mostly just **addition, subtraction, multiplication, and division**:

$$+, \ -, \ \times, \ \div$$

- This is the same **grade school arithmetic** used to add up a grocery bill. The difference is that GPT performs these operations an enormous number of times in a row.
- Nothing more mysterious than these four operations is needed to turn a row of word vectors into a list of next-word probabilities.
- The **knowledge** the system uses to transform the x's into the pdf is stored **in a bunch of numbers called w's, or weights**.
    - The x's change every time a new sentence comes in, but the w's stay fixed while the system is being used. The arithmetic combines the incoming x's with the stored w's, and the result is the pdf.
    - The knowledge is located **in** the weights. If the weights are changed, the system's predictions change, which means its knowledge has changed.
- Today's AI systems have **about a couple of trillion weights**. Every piece of knowledge the system has, including the fact that ducks quack and chickens cackle, is spread across those trillions of numbers.

Section 11: Weights and Synapses

- A couple of trillion weights may seem like far too many numbers for the system to resemble human knowledge. The comparison with the brain points the other way.
- A person knows that chickens cackle but ducks quack, and that knowledge has to be stored somewhere. It is stored in the **synapses**, which are the connections between the **neurons** of the brain. A synapse is the point where one neuron passes a signal to another, and the strength of each connection helps determine what the brain does with incoming information.
- The human brain has **about 150 trillion synapses**. Comparing this to the roughly 2 trillion weights of an AI system gives:

$$\frac{150 \text{ trillion}}{2 \text{ trillion}} = 75$$

- The brain therefore has roughly 75 times as many connections as today's largest AI systems have weights.
- In both cases, knowledge is stored in a huge number of connection strengths rather than in a list of written facts. The weights of an AI system play the role that synapses play in the brain, so the number of weights is not unreasonably large when compared to human knowledge.

Section 12: Knowledge as Completion, or Prediction

- The claim that **knowledge is completion** strikes many people as nonsensical. Among them is **Yann LeCun**, a leading AI researcher, who argues that this view does not make sense and just raised **a billion dollars** on the basis of that position.
- Anyone who has used an AI system can judge the question directly. The question is whether, while interacting with the system, it feels like a system whose notion of knowledge is **totally different** from our own.
    - One reasonable answer is that whatever the system's notion of knowledge is, it is **not totally different** from ours. Its answers, completions, and explanations are close enough to human knowledge that the two cannot be entirely separate.
- Another term for completion is **prediction**, and that word tends to sit better with people. Humans constantly **predict** and **expect** what will happen next, whether it is the next word in a sentence, the next note in a song, or the next event in a situation. Seen this way, knowledge as completion is knowledge as prediction, which is something human minds do all the time.

Section 13: Learning by Making Mistakes

- One of the most central issues in AI is **how it learns**. AI **learns by making mistakes**, and humans may not be that different.
- The cartoon titled "How much you learn," taken from the r/GetMotivated page on Reddit, shows three stacks of books sitting side by side.
    - The stack labeled **"From theory"** is the shortest, with only a few red books.
    - The stack labeled **"From practice"** is taller, with the red books plus several blue books on top.
    - The stack labeled **"From mistakes"** is by far the tallest, containing red and blue books topped by a large pile of green books.
    - The cartoon expresses the idea that people learn the most from their mistakes, which matches the way AI systems learn.
- The main algorithm for learning in AI is called **Stochastic Gradient Descent (SGD)**. The algorithm can be understood through a very easy example: **averaging**.

Section 14: Averaging as Learning

- Computing an average in the ordinary way is simple. The average of 10, 12, and 14 is found by adding the numbers and dividing by how many there are:

$$\frac{10 + 12 + 14}{3} = \frac{36}{3} = 12$$

- This direct method **does not work** for the complex things AI deals with, such as **speech, language, biological molecules, and scenes**. For those, there is no simple formula that can be applied once to get the answer, so a different approach is needed.
- The averaging method used in AI works in a way that can be extended to those complex cases.
    - We start with a **list of numbers**.
    - We make a **wild guess**, called x, about what their average is.
    - We then **improve x through exposure to the data**. We take one number from the list and compare it to the current value of x. We then move x **a bit closer** to that observed number.
    - We **repeat** this with another number, and then another.
- This is called **observation-based error learning**. The estimate starts out wrong, each observation reveals how wrong it is, and the estimate is corrected a little each time.

Section 15: Stochastic Gradient Descent, Step by Step

- The goal is to find the average of the list [78, 67, 98, 85, 76, 68, 98, 72]. The average can be thought of as **the essence** of the list, a single number that best represents all the values in it.
    - For reference, the true average of this list can be computed directly:

$$78 + 67 + 98 + 85 + 76 + 68 + 98 + 72 = 642$$

$$\frac{642}{8} = 80.25$$

- SGD reaches an estimate close to this value without ever doing that full calculation. It follows these steps:
    1. **Pick any random number x as the estimate of the average.** This is **time 0** in the algorithm, so the estimate is written as x[0]. Suppose the random number picked is 130, so x[0] = 130. The guess is deliberately far from the true value, and the algorithm improves it by observation.
    2. **Randomly pick a number d from the data.** This shows how far the data is from the estimate, and the average should be close to the data. Suppose the number picked is 98, so d[0] = 98.
    3. **Measure the mistake.** The mistake is called the **Loss**, and it is the difference between the current estimate and the observed data point:

$$\text{Loss} = x[0] - d[0] = 130 - 98 = 32$$

    4. **Improve the estimate by subtracting a small part of the Loss from it.** Here that small part is 10% of the Loss:

$$x[1] = x[0] - 10\% \text{ of } 32$$

$$10\% \text{ of } 32 = 0.10 \times 32 = 3.2$$

$$x[1] = 130 - 3.2 = 126.8$$

    5. **Repeat.** Pick another number d from the data and use it in the same way to improve x[1] into x[2], then x[2] into x[3], and so on.

- The same rule works when the estimate starts too low. If the estimate were 10 and the data point drawn were 78, the Loss would be 10 minus 78, which is negative 68. Subtracting 10% of a negative number means adding, so the estimate would rise by 6.8, moving from 10 up to 16.8. The estimate is always pushed toward the data, whichever side it starts on.
- The whole algorithm does two things. It **makes an initial estimate or guess**, and it **uses data to improve that estimate**.
- Each part of the name describes one part of the method.
    - **Stochastic** means **random**, since each data point is picked at random from the list.

- **Gradient** refers to the **Loss**, since the size and direction of each correction depend on how large the mistake is and which way it points.
- **Descent** means **reducing**, since each step reduces the mistake by moving the estimate closer to the data.

Section 16: The Effect of the Learning Rate

- The fraction of the Loss subtracted at each step is called the **learning rate**. In the example it was 10%. The four graphs show SGD applied to the list above with four different learning rates: **10%, 50%, 1%, and 80%**.
  - In every graph, the vertical axis is the estimate x, running from 0 to 140, and the horizontal axis is **epochs**, running from 0 to 300. Each epoch here is one update step of the estimate.
  - Each graph shows three runs that start from three different **initial conditions (IC)**, meaning three different starting guesses. The red run starts at **IC = 130**, the green run starts at **IC = 80**, and the blue run starts at **IC = 10**.
- With a learning rate of **10%**, the red run drops steeply from 130 and the blue run climbs steeply from 10. Both reach the neighborhood of 80 within roughly the first 25 to 50 epochs. The green run starts near 80 and stays there. After that, all three runs overlap and wobble slightly around 80, staying within a narrow band a few points above and below it.
- With a learning rate of **1%**, all three runs are **smooth**, with almost no wobble. However, learning is **very slow**. After 300 epochs, the red run has only come down to about 82 and the blue run has only climbed to about 76, so neither has fully reached the average yet. The green run stays flat near 80 the whole time.
- With a learning rate of **50%**, the red and blue runs jump to the neighborhood of 80 within the first few epochs, so learning is very fast. Afterward, however, all three runs **bounce up and down widely**, roughly between 70 and 95, and never settle on a single value.
- With a learning rate of **80%**, the runs reach the neighborhood of 80 even faster, but the bouncing is **even larger**, spreading roughly between 65 and 97.
- The bouncing comes from the random single data points used at each step. Each update pulls the estimate toward just one number from the list, and those numbers range from 67 to 98. With a small learning rate, each pull moves the estimate only slightly, so the pulls toward high and low values average out. With a large learning rate, each pull moves the estimate most of the way toward whichever number was drawn. At 80%, if the estimate is 80 and the number drawn is 98, the estimate jumps by 80% of the 18-point gap, which is 14.4, landing at 94.4. The next draw may then throw it far back down.

Section 17: The Fundamental Learning Tradeoff

- The four graphs reveal the **fundamental learning tradeoff**. The learning rate is written with the Greek letter γ (gamma).
  - When γ is **increased**, learning **speed increases**, since the estimate reaches the neighborhood of the answer in just a few steps. At the same time, **accuracy decreases**, since the estimate keeps jumping around instead of settling close to the true average.
  - When γ is **decreased**, learning **speed decreases**, since the estimate creeps toward the answer and may still be far from it after hundreds of steps. At the same time, **accuracy increases**, since the estimate moves smoothly and stays close to the true value once it gets there.
- Speed and accuracy pull against each other, so a single learning rate cannot make both as good as possible at once. The 10% learning rate sits between the extremes, reaching the average fairly quickly while wobbling only a little.
- For generations, **thousands of people have made their living** by finding better ways to **optimize** learning. Optimizing here means adjusting how learning is carried out so that **speed and accuracy both increase** at the same time, getting around the tradeoff that a single fixed learning rate cannot escape.

Now for the lecture. first analyze the entire lecture and extract all informatio n from the text, images, fi gures, graphs, and diagra ms internally. DO NOT P...
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Section 1: Watching Speech Inside the Vocal Tract

- **Real-time MRI** (magnetic resonance imaging) films the inside of the head while a person talks, with the sound recorded at the same time. The images are **midsagittal**, meaning a side-view slice straight down the middle of the head, with the face on the left and the spinal column as the bright vertical band on the right.
    - Air gives off almost no signal, so the open airways appear black: the mouth cavity, the nasal passages above the roof of the mouth, and the **pharynx**, the air tube at the back of the throat behind the tongue.
    - The large gray mass filling the mouth is the **tongue**. Above it is the **hard palate**, the bony roof of the mouth, and hanging from its back edge is the **velum**, or soft palate, which rises to close the passage into the nose and lowers to open it.
- During speech, the tongue, lips, jaw, and velum all move at once and very quickly. Each speech sound is produced by shaping this tube of air in a particular way, and **phonetics** is the study of these speech sounds.

Section 2: Fricatives

- A **fricative** is a consonant made by bringing two parts of the mouth very close together without fully closing them. Air forced through the narrow gap becomes turbulent and produces a hissing noise, as in f and s.
- English fricatives differ in **place of articulation**, which is where the narrow gap is formed, and **voicing**, which is whether the **vocal folds** in the larynx vibrate during the sound.
    - A **voiceless** sound is made with the vocal folds not vibrating, so the only noise is the hiss of air.
    - A **voiced** sound is made with the vocal folds vibrating, so the hiss is combined with a buzz.
- The English fricatives form **four places**, each with a voiceless and a voiced member.
    - **Labial**: f (voiceless) and v (voiced). The lower lip rises toward the upper teeth while the tongue stays low in the mouth.
    - **Dental**: voiceless th (as in "think") and voiced th (as in "this"). The front of the tongue is pushed forward to the teeth.
    - **Alveolar**: s (voiceless) and z (voiced). The tongue tip is raised toward the **alveolar ridge**, the bony ridge just behind the upper front teeth.
    - **Postalveolar**: sh (voiceless) and its voiced partner j. The narrowing is just behind the alveolar ridge, slightly farther back than for s and z, with more of the front of the tongue raised toward the roof of the mouth.
- The members of each pair share the **same place and the same manner**. The only difference between f and v, between the two th sounds, between s and z, and between sh and j is whether the vocal folds are vibrating.

Section 3: Stops

- A **stop** is a consonant made by closing the mouth completely at some point. English has two kinds.
    - In an **oral stop**, also called a **plosive**, the velum is raised and seals off the nose. With both mouth and nose closed, air pressure builds behind the closure and bursts out when the closure is released.
    - In a **nasal stop**, the mouth is closed in the same way, but the velum is lowered, so air flows out through the nose the whole time.

- The stops form **three places of articulation**, each with a voiceless oral stop, a voiced oral stop, and a nasal stop.
    - At the **lips**: p (voiceless), b (voiced), and m (nasal). The two lips are pressed together.
    - At the **alveolar ridge**: t (voiceless), d (voiced), and n (nasal). The tongue tip presses against the ridge behind the upper teeth.
    - At the **velum**: k (voiceless), g (voiced), and ng (nasal), the sound at the end of "sing." A sound made here is called **velar**, and the back of the tongue humps up against the soft palate.
- In m, n, and ng, the velum hangs down away from the back of the throat, leaving the passage to the nose open. In p, t, and k, the velum is raised and the passage is sealed. In real time, each stop unfolds as a closure forming, then building pressure or nasal airflow, then releasing.

Section 4: The Structure of English Consonants

- The fricatives and stops fit into one table organized by three properties: **manner of articulation**, **place of articulation**, and **voice**. Manner is how the air is shaped: a narrow gap for a fricative, a full closure with the nose sealed for an oral stop, and a full closure with the nose open for a nasal stop.
    - The places run from front to back: **labial**, **dental**, **alveolar**, **postalveolar**, and **velar**.
    - **Fricatives**: voiceless f, th, s, sh and voiced v, th, z, j, filling the labial, dental, alveolar, and postalveolar columns.
    - **Oral stops (plosives)**: voiceless p, t, k and voiced b, d, g, filling the labial, alveolar, and velar columns.
    - **Nasal stops**: m, n, and ng, all voiced, filling the labial, alveolar, and velar columns.
- Every consonant in the table is defined by one value of each property. For example, t is a **voiceless alveolar oral stop**. Changing only its voicing gives d, and changing d from an oral stop to a nasal stop gives n.
- The table has no dental or postalveolar stops and no velar fricatives. English speakers use this entire system every time they talk without ever being taught it, which shows how much **hidden structure** exists in a speaker's knowledge of English.

Section 5: What AI Is

- **Artificial intelligence (AI)** is a model of several **psychological skills that animals have**: **knowing**, **recognizing**, **attending**, **acting**, and **learning**.
- The way AI models each of these skills is **at least slightly based on how humans perform them**. AI is not a technical innovation that came out of nowhere; each of its abilities was built as a model of a human skill.

Section 6: Taylor's Cloze Procedure

- AI's way of knowing traces back to a 1953 paper by **Wilson L. Taylor**, **"Cloze Procedure": A New Tool for Measuring Readability**, published in **Journalism Quarterly** in Fall 1953. The journal was devoted to research studies in the field of mass communications.
    - The paper was the first comprehensive statement of the method and its theory, which had been introduced briefly during a workshop at the **1953 AEJ convention**.
    - It reported findings from **three pilot studies and two experiments** comparing cloze procedure results with those of two **readability formulas**, which estimate how easy a piece of writing is to read.
- Taylor described cloze procedure as a new **psychological tool for measuring the effectiveness of communication**. The method is straightforward, its data are easily turned into numbers, and its findings held up.
    - It was first treated as a new approach to **readability**. Its results repeatedly agreed with two standard readability devices, the **Flesch** and **Dale-Chall** formulas, and it was then tested directly against those formulas.

    - If future research supported the results, Taylor expected many theoretical and practical uses in other fields involving communication.
- Taylor credited two experimental psychologists, **Prof. Charles E. Osgood** of the University of Illinois and **Melvin R. Marks** of the Personnel Research Section, A.G.O., Department of the Army, for starting and assisting the work that produced cloze procedure.
    - Other contributors were **Prof. Lee J. Cronbach**, an educational psychologist and statistician; **Dean Wilbur Schramm** of the Division of Communications; **Prof. Charles E. Swanson** of the Institute of Communications Research; and psychologist **George R. Klare**. Swanson and Klare had both written on readability.
    - **Kalmer E. Stordahl** and **Clifford M. Christensen**, research associates of the Institute, also contributed, along with journalism teachers who lent their classes.

Section 7: Closure and the Cloze Unit

- The unit at the center of Taylor's method is the **cloze**, pronounced like the verb "close" and taken from the word **closure**.
    - **Closure** is a term from **Gestalt psychology** for the human tendency to **complete a familiar but not-quite-finished pattern**. We see a broken circle as a whole circle by mentally closing the gap.
    - Three black circles, each missing a wedge, with the wedges pointing toward a common center, are seen as a white triangle lying on top of them. No triangle is actually drawn; the visual system completes it from the partial edges.
- A broken circle can be completed because its shape is **so familiar** that it is recognized even when much of it is missing. The same principle applies to language. Given "Chickens cackle and ___ quack," almost anyone instantly supplies "ducks." If that word matches the one removed, the person scores **one cloze unit** for correctly closing the gap.
- Filling that one blank requires knowing many smaller patterns at once.
    - A reader must know the **meanings** of all five words, which are patterns of relationships between symbols and meanings, and their **forms**, which are patterns of letters.
    - A reader must also know the meanings of **combinations** of the words, and recognize that the sentence structure calls for a parallel: an animal that goes with "quack," just as chickens go with "cackle."
    - The reader must guess what the incomplete sentence means as a whole and then complete its pattern to fit that whole meaning.
- Taylor gave two definitions.
    - A **cloze unit** is any single occurrence of a **successful attempt to accurately reproduce a part deleted from a "message"** (any language product) by deciding from the remaining context what the missing part should be.
    - **Cloze procedure** is a method of **intercepting a message from a "transmitter"** (a writer or speaker), **mutilating its language patterns by deleting parts**, and giving it to **"receivers"** (readers or listeners), whose attempts to make the patterns whole again produce a countable number of cloze units.

Section 8: How AI Knows

- AI shows that it knows something by **demonstrating the ability to complete, or cloze, sequences**.
- A person's knowledge of a skill can be tested in the same way, by asking them to complete a sequence and checking whether the completion is correct.
- In AI, **knowledge is nothing more than completion**. A system knows English if, given part of an English sequence, it can correctly produce what is missing.

Section 9: How GPT Represents Words

- **GPT** is given a sequence of words and predicts the next word. To a computer, a word is just **a bunch of random numbers**, and the only requirement is that **no two words are given the same numbers**.

- A list of numbers is called a **vector**, and a word's vector could contain **1024 numbers**. The input "chickens cackle and ducks" is therefore four vectors, one per word.
- If the system **knows English**, it outputs a single vector of **10,000 numbers**, where each number is the **probability that one particular English word is the next word**.
    - For "chickens cackle and ducks ___," "quack" receives a high probability, while words such as "a," "cat," and "Zulu" receive low probabilities.
    - Predicting the next word is a cloze task, filling in a blank at the end of a sequence.

Section 10: How GPT Transforms Inputs into Predictions

- The output list of probabilities is a **pdf** (probability distribution function), and the input vectors are called the **x's**. GPT turns the x's into the pdf **through recognizing by computing**.
- The computing is mostly **grade school arithmetic**:

$$+,\ -,\ \times,\ \div$$

- The **knowledge** used to transform the x's into the pdf is stored **in** a bunch of numbers called **w's**, or **weights**. The arithmetic combines the incoming x's with the stored w's to produce the pdf, so changing the weights changes the predictions.
- Today's AI systems have **about a couple of trillion weights**.

Section 11: Weights and Synapses

- A couple of trillion weights may seem like too many to resemble human knowledge. However, human knowledge, such as knowing that chickens cackle but ducks quack, is stored in the **synapses**, the connections between the **neurons** of the brain.
- The human brain has **about 150 trillion synapses**:

$$\frac{150 \text{ trillion}}{2 \text{ trillion}} = 75$$

- The brain has roughly 75 times as many synapses as an AI system has weights. In both systems, knowledge is stored in a very large number of connection values, so trillions of weights are not excessive compared to the brain.

Section 12: Knowledge as Completion, or Prediction

- Many people consider the idea that knowledge is completion to be nonsense. **Yann LeCun** takes this position and just raised **a billion dollars** on the basis that it does not make sense.
- The question can be judged directly: when interacting with an AI system, it is worth asking whether its notion of knowledge seems **totally different** from our own. One answer is that whatever its notion of knowledge is, it is **not totally different** from a human's.
- Another term for completion is **prediction**, a word that sits better with people. Humans also **predict** and **expect** what comes next.

Section 13: Learning by Making Mistakes

- One of the most central issues in AI is **how it learns**, and AI **learns by making mistakes**. Humans may not be that different.
    - The cartoon "How much you learn," from a Reddit motivational page (r/GetMotivated), compares three stacks of books: a short stack learned **from theory**, a taller stack **from practice**, and the tallest stack **from mistakes**.
- The main learning algorithm in AI is **Stochastic Gradient Descent (SGD)**, and its logic can be shown with **averaging**.

Section 14: Averaging as Learning

- The average of 10, 12, and 14 is found directly:

$$\frac{10+12+14}{3}=\frac{36}{3}=12$$

- This direct method **does not work** for the complex things AI handles, such as **speech, language, biological molecules, and scenes**.
- The AI averaging method works differently. We start with a list of numbers and make a **wild guess x** about their average. We then take one number from the list, compare it to the current x, and move x **a bit closer** to that number. This is repeated with another number, and then another.
- This is **observation-based error learning**: the estimate is improved each time it is compared with an observed piece of data.

Section 15: Stochastic Gradient Descent, Step by Step

- The goal is to find the average, **the essence**, of [78, 67, 98, 85, 76, 68, 98, 72]. Computed directly, the true average is:

$$78+67+98+85+76+68+98+72=642$$

$$\frac{642}{8}=80.25$$

- SGD estimates this value through the following steps.
  1. **Pick any random number x as the estimate of the average.** This is **time 0**, so the estimate is x[0]. If the number picked is 130, then x[0] = 130.
  2. **Randomly pick a number d from the data** to see how far the data is from the estimate, since the average should be close to the data. If the number picked is 98, then d[0] = 98.
  3. **Measure the mistake, called the Loss:**

$$\text{Loss}=130-98=32$$

4. **Improve the estimate by subtracting a bit of the Loss, here 10%, from the current estimate:**

$$x[1]=x[0]-10\%\text{ of }32$$

$$x[1]=130-3.2=126.8$$

5. **Repeat.** Pick another number d from the data and use it to improve x[1] into x[2], and so on.

- When the estimate is below the data point, the Loss is negative, and subtracting part of a negative Loss raises the estimate. For an estimate of 10 and a data point of 78, the Loss is negative 68, so the estimate rises by 6.8 to 16.8. Either way, each step moves the estimate toward the data.
- The algorithm does two things: it **makes an initial estimate**, and it **uses data to improve that estimate**. The name describes the method: **Stochastic** means **random**, since data points are picked at random; **Gradient** refers to the **Loss**; and **Descent** means **reduce**, since each step reduces the Loss.

Section 16: The Effect of the Learning Rate

- The fraction of the Loss subtracted at each step is the **learning rate**, which was 10% in the example. SGD on the same list was run for 300 epochs at learning rates of **10%, 50%, 1%, and 80%**, each epoch being one update of the estimate. Each rate was run from three **initial conditions (IC)**, meaning starting guesses: **IC = 130**, **IC = 80**, and **IC = 10**.
- At **10%**, the estimates starting at 130 and 10 reach about 80 within roughly 25 to 50 epochs, and the estimate starting at 80 stays near 80. All three then wobble slightly around 80.
- At **1%**, all three estimates move smoothly with almost no wobble, but very slowly. After 300 epochs, the estimate from 130 has only fallen to about 82 and the estimate from 10 has only risen to about 76.

- At **50%**, the estimates reach the neighborhood of 80 within a few epochs but then swing widely, roughly between 70 and 95, never settling.
- At **80%**, the estimates reach 80 even faster, but the swings are larger still, roughly between 65 and 97.
- The swinging comes from updating on one randomly drawn data point at a time, with the data ranging from 67 to 98. A large learning rate moves the estimate most of the way toward whichever point is drawn. At 80%, an estimate of 80 that draws 98 moves by 80% of the 18-point gap, which is 14.4, landing at 94.4.

Section 17: The Fundamental Learning Tradeoff

- The learning rate is written as γ (gamma), and it controls a **fundamental learning tradeoff** between speed and accuracy.
    - When γ increases, learning **speed increases**, but **accuracy decreases**, since the estimate keeps jumping around the true value.
    - When γ decreases, **accuracy increases**, since the estimate moves smoothly, but learning **speed decreases**, since it takes many more steps to reach the true value.
- For generations, **thousands of people have made their living** finding better ways to **optimize** learning so that **speed and accuracy both increase** at the same time.